

BEYOND WORDS AI
AI POWERED HAND GESTURE RECOGNITION AND
CHATBOT FOR ASSISTIVE COMMUNICATION
A PROJECT REPORT

Submitted by

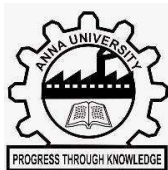
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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY: CHENNAI 600 025

OCTOBER 2026

**RAJALAKSHMI ENGINEERING COLLEGE
CHENNAI**

BONAFIDE CERTIFICATE

Certified that this Project report titled “NITROSIS” is the bonafide work of
“**BALAMURUGAN J (221401017), MEDVIN DEVA I R (221401062)**” who carried
out the work under my supervision. Certified further that to the best of my knowledge the
work reported herein does not form part of any other project report or dissertation on the
basis of which a degree or award was conferred on an earlier occasion on this or any other
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Submitted to Project Viva-Voce Examination held on

Internal Examiner

External Examiner

RAJALAKSHMI ENGINEERING COLLEGE
B.Tech. COMPUTER SCIENCE AND BUSINESS SYSTEMS

VISION

- 3 To promote highly ethical and innovative computer professionals through excellence in teaching, training and research.

MISSION

- 4 To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.
- 5 To promote research activities amongst the students and the members of faculty that could benefit the society.
- 6 To impart moral and ethical values in their profession

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- To equip students with state-of-the-art contemporary technology in computer science and basic engineering science.
- To expose students to the business principles aligned with their domain.
- To educate on the service orientation principles for various business disciplines.
- To encourage innovative research through creative thinking and critical analytical ability.
- To inculcate ethical and moral values in the individuals benefitting the society.

PROGRAM OUTCOMES (POs)

A graduate of the Computer Science and Business Systems program will demonstrate:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Computer Science and Business Systems program will demonstrate:

- 1 The students graduating will have sound knowledge in Computer Science with equal appreciation of humanities, management sciences and human values.
- 2 The students will explore emerging topics such as Analytics, Machine Learning, Cloud Computing, and Internet of Things and apply the knowledge to design and develop solutions to societal needs.
- 3 The students will be industry ready with required business skills in service orientation.

OBJECTIVES:

1. To identify a problem statement of societal importance.
2. To analyze a problem and find out requirements.
3. To plan, design, and execute a project solution effectively within specified timelines and resources.
4. To develop technical expertise in using contemporary tools, technologies, and methodologies for the project.
5. To test and document a project.

COURSE OUTCOMES (COs)

CO1	Apply the concepts of computer science to real-world problems and understand Ethical and societal implications.
CO2	Take in any difficult sensible issues and propose solution through formulating right methodology.
CO3	Attain a hands-on experience in changing a small novel idea / method right into an operating model / prototype related to multi-disciplinary abilities and / or understanding and operating in at team
CO4	Interpret the outcome of their project. Students will take on the challenges of teamwork, prepare a presentation in a professional manner, and document all aspects of design work.
CO5	Publish or release the project outcome to the society.

CB23721–Project Evaluation I

Project Title: BEYOND WORDS AI

Batch Members: BALAMURUGAN J(221401017) ,

MEDVIN DEVA IR (221401062)

Name of the Supervisor: Dr. P.Hemalatha

CO-PO-PSO MAPPING:

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	2	1	1	3	1	2	2	-	1		2	2	2	1
CO2	3	3	2	2	3	-	1	1	-	1		2	3	3	2
CO3	3	2	3	2	3	-	1	1	2	2		1	3	3	3
CO4	1	1	2	2	2	-	-	-	3	3		1	2	2	2
CO5	1	1	1	1	2	-	-	-	2	3	3	2	2	3	3

Signature of the Supervisor



CO - SDG Relevance Record
B.Tech - Computer Science and Business Systems

Regulation : 2019

Semester : VII

Course Code & Name : CB23721- Project Evaluation I

SDG No. & Theme	Addressing CO	Topic / Activity addressing SDG Theme
SDG 9 – Industry, Innovation and Infrastructure	CO1,CO2,CO3,CO4	The system combines IoT, AI, cloud, and mobile technologies to automate farming and improve efficiency.
SDG 12 – Responsible Consumption and Production	CO2,CO3,CO5	Natural fertilizer recycling, reduced chemical fertilizer use, sustainable crop rotation techniques
SDG 15 – Life on Land	CO3,CO4,CO5	Uses chemical-free pest control and aquaponics to maintain soil health and reduce environmental impact.

Signature of the Supervisor

Signature of the HoD

ABSTRACT

Communication barriers can create significant challenges for individuals who use sign language, particularly when interacting with people who do not understand sign language. Beyond Words is an AI-based sign-language communication system designed to reduce this barrier by recognizing and interpreting sign-language gestures and converting them into understandable text or speech. The system is also extended with a sign-language-driven AI chatbot so that recognized sign-language text can be used as conversational input without requiring typed queries.

The proposed system applies Artificial Intelligence and computer vision techniques to identify hand gestures and sign-language patterns from visual input. The recognized signs are processed and mapped to their corresponding meanings, allowing the system to generate an appropriate textual or spoken output. The system is designed with a user-friendly interface to make communication more convenient and accessible. After sign recognition and text conversion, the recognized text can be passed to an AI chatbot module, which processes the user query and generates a conversational response that can be presented as text and, where required, speech.

A review of existing research on sign-language recognition, gesture recognition, computer vision, and AI-based communication systems was carried out to identify existing approaches, technologies, and limitations. Based on the identified research gap, Beyond Words focuses on developing an accessible solution that can support real-time sign-language interpretation and improve communication efficiency.

The project demonstrates the potential of AI to bridge communication gaps and contribute towards more inclusive digital communication. The system focuses on recognizing predefined gestures and providing understandable output through text and speech. The chatbot extension further supports interactive communication by allowing sign-language users to communicate with an AI system through recognized gestures.

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CHAPTER 1

INTRODUCTION

1. GENERAL

Communication is an essential part of everyday human interaction, enabling individuals to express their thoughts, needs, emotions, and ideas. However, people who communicate primarily through sign language may face difficulties when interacting with individuals who do not understand or use sign language. This communication gap can create challenges in education, healthcare, employment, public services, and social interaction. With the rapid development of Artificial Intelligence, Machine Learning, and Computer Vision, technology can be used to reduce these communication barriers. Sign-language recognition systems can analyze visual information such as hand gestures and movements and convert them into meaningful text or speech. Beyond Words is proposed as an AI-based sign-language communication system that aims to bridge the communication gap between sign-language users and non-sign-language users.

The system uses computer vision and machine-learning techniques to recognize sign-language gestures from visual input and interpret them into understandable output. It focuses on creating a simple and accessible interface through which users can communicate more effectively. By combining sign-language recognition with modern AI techniques, Beyond Words aims to demonstrate how intelligent systems can support more natural, efficient, and inclusive communication.

The project is motivated by the need for inclusive communication technologies that can assist people in overcoming language and accessibility barriers. The system is intended to recognize predefined gestures, map them to their meanings, and provide text or speech output to support communication.

1.1 Problem Statement

People who use sign language may face communication difficulties when interacting with individuals who do not understand sign language. Existing recognition systems can be limited by vocabulary, difficulty with dynamic gestures, environmental conditions, user variation, computational requirements, and the absence of speech output.

There is therefore a need for an accessible AI-based system that can recognize predefined sign-language gestures from visual input and convert them into understandable text or speech. Beyond Words addresses this need by combining gesture recognition, AI-based classification, and accessible output in a simple communication interface.

1.2 OBJECTIVES

The primary objective of Beyond Words is to develop an AI-based system that can recognize and interpret sign-language gestures and convert them into understandable text or speech, thereby reducing communication barriers between sign-language users and non-sign-language users.

The specific objectives are:

1. To integrate an AI chatbot that accepts recognized sign-language text as user input and generates conversational responses.
2. To develop an AI-based sign-language recognition system capable of identifying predefined hand gestures from visual input.
3. To apply computer vision and machine-learning techniques for accurate detection and classification of sign-language gestures.
4. To convert recognized signs into meaningful text that can be easily understood by the receiving user.
5. To provide speech output from the recognized signs to support more natural communication.
6. To develop a simple and user-friendly interface that can be used without extensive technical knowledge.

1.3 SCOPE OF THE PROJECT

The scope of Beyond Words is to develop an AI-based sign-language communication system that recognizes predefined gestures from camera input and converts them into understandable text and speech. The system focuses on accessible, real-time interaction rather than complete sign-language vocabulary coverage. The proposed scope is also extended to include conversational interaction with an AI chatbot using recognized sign-language text as input.

The system covers:

1. Recognition of predefined hand gestures captured through a camera.
2. Computer-vision-based detection and machine-learning-based classification of sign-language gestures.
3. Conversion of recognized gestures into meaningful textual output.
4. Generation of speech output from the recognized text.
5. Real-time processing with a simple and user-friendly interface.
6. Support for communication between sign-language users and people who do not understand sign language.
7. Interaction with an AI chatbot using recognized sign-language text as conversational input.
8. Consideration of practical challenges such as lighting, background, camera angle, signing speed, and differences between users.
9. Study of existing sign-language recognition approaches and their limitations as a basis for improving accessibility.

1.4 PROPOSED SYSTEM AND CORE ADVANTAGES

Beyond Words is an AI-based sign-language communication system proposed to reduce the communication gap between sign-language users and people who do not understand sign language. The system receives visual input through a camera, processes hand gestures, identifies the corresponding sign, and maps the recognized gesture to its associated meaning. In addition to direct text and speech output, the system includes a sign-language-driven AI chatbot extension. Once a gesture is recognized and converted into text, the interpreted text can be sent to the chatbot as a query, allowing the user to interact without typing.

The recognized gesture is displayed as text and can also be converted into speech, enabling the non-sign-language user to understand the intended message without requiring prior knowledge of sign language. The proposed system focuses on a simple and accessible interface where users can perform signs naturally in front of a camera.

Key System Advantages:

1. Reduces communication barriers: Helps sign-language users communicate more effectively with people who do not understand sign language.
2. AI-based and real-time recognition: Uses Artificial Intelligence, computer vision, and machine-learning techniques to recognize and interpret predefined sign-language gestures.
3. Text and speech output: Converts recognized signs into text and can provide speech output, making the interpreted message easier for non-sign-language users to understand.
4. AI chatbot interaction: Allows recognized sign-language text to be used as input to a conversational AI chatbot, with the generated response presented as text and optionally speech.

CHAPTER 2

LITERATURE REVIEW

2.1 OVERVIEW OF RELATED WORK

Recent research has explored Artificial Intelligence, Machine Learning, Computer Vision, and Deep Learning for sign-language recognition and gesture-based communication. Traditional systems generally capture hand images or video, preprocess the input, extract relevant features, and classify the gesture using machine-learning algorithms.

Research has also investigated deep-learning techniques for image-based and video-based recognition, as well as hand-landmark and keypoint approaches for identifying hand position, orientation, movement, and finger configuration. These methods improve recognition capability but remain sensitive to dataset size, environmental conditions, and signer variation.

Sign-to-text and sign-to-speech systems extend recognition into practical communication by presenting the interpreted sign as written or spoken output. The reviewed work shows that challenges remain in dynamic gesture recognition, real-time processing, vocabulary coverage, environmental robustness, and user-independent performance. Beyond Words combines recognition with text and speech output to address these communication needs.

2.2 COMPARATIVE ANALYSIS

The I literature table compares different sign-language recognition and communication approaches, including neural machine translation, keypoint estimation, deep learning, MediaPipe-based landmark recognition, transfer learning, temporal sequence modelling, sign-to-speech systems, and multimodal sign-language processing. The comparison identifies reported performance measures and limitations relevant to the proposed Beyond Words system.

Author	Name	Methodology	Performance Measure	Limitations
Wesley F. Maia et al. (2025)	<i>Automatic Sign Language to Text Translation Using MediaPipe and Transformer Architectures</i>	Uses MediaPipe/body keypoints , convolutional keypoint embeddings, positional encoding and Transformer-based sequence modelling for Sign-to-Gloss and Gloss-to-Text translation.	Evaluated using sign-language translation metrics such as BLEU and compared against baseline translation models. The authors report comparable translation performance while using body keypoints. (ScienceDirect)	Uses an intermediate gloss representation , and continuous sign-language translation remains challenging. Performance also depends on reliable body-keypoint extraction.
Ryan Wong, Necati Cihan Camgoz & Richard Bowden (2024)	<i>Sign2GPT: Leveraging Large Language Models for Gloss-Free Sign Language Translation</i>	Uses visual sign-language representations with a large language model (LLM) to perform gloss-free sign-language translation , reducing dependence on manually annotated glosses.	Evaluated using machine-translation metrics such as BLEU on sign-language translation datasets.	Requires substantial training data and computational resources; gloss-free translation remains difficult because sign language contains complex temporal and linguistic information.
Jia Gong et al. (2024)	<i>LLMs are Good Sign Language Translators</i>	Introduces SignLLM , integrating sign-language visual representations with large language models to translate sign sequences into natural language.	Evaluated using sign-language translation benchmarks and automatic translation metrics , demonstrating the effectiveness of LLM-based translation.	LLM-based approaches can require significant computational resources and depend on the quality and diversity of sign-language training data.

Youngjoon Jang et al. (2025)	<i>Lost in Translation, Found in Context: Sign Language Translation with Contextual Cues</i>	Combines visual sign features with background captions, previous-sentence translations and pseudo-glosses , feeding these contextual cues into a pretrained LLM. Evaluated on BOBSL and How2Sign .	Reports improved translation quality over previously reported results on BOBSL and competitive results on How2Sign. (CVF Open Access)	Requires additional contextual information and multiple processing components. The approach is also evaluated primarily on BSL/ASL datasets , rather than Indian Sign Language.
Shagun Katoch, Varsha Singh & Uma Shanker Tiwary (2022)	<i>Indian Sign Language Recognition System Using SURF with SVM and CNN</i>	Uses Bag of Visual Words, SURF feature extraction, SVM and CNN to recognize 26 alphabets and 10 digits from live video. Includes text and speech output.	CNN achieved 99.64% test accuracy , while SVM achieved 99.17% . CNN also achieved 99.57% precision, recall and F1-score. (DOI)	Mainly focuses on isolated alphabets and digits rather than continuous sign-language sentences. Performance can depend on camera/background conditions.
Chandrakala Arya et al. (2024)	<i>A Lightweight Solution for Real-Time Indian Sign Language Recognition</i>	Uses YOLO-based object detection for real-time recognition of Indian Sign Language letters and numbers. The study compares different YOLO versions.	Reported 98.46% precision, 98.78% recall and 97.54% mAP@0.5 for the proposed approach. (IEEE Xplore)	Primarily addresses letters and numbers , rather than continuous sentence-level translation. Performance depends on the training dataset and real-world deployment conditions.
Lincoln Amory Rane et al. (2025)	<i>Khasi Sign Language Recognition Using Google's MediaPipe and Deep Learning Feedforward Neural Network Approach</i>	Uses Google MediaPipe to extract hand/body landmark information and a deep-learning feedforward neural network for regional sign-language recognition.	Evaluates the model using classification accuracy and related performance metrics on the Khasi Sign Language dataset.	Focuses on a regional sign language and limited vocabulary ; results may not directly generalize to Indian Sign Language or continuous translation.
Shilpa Ingole & Jagdish Bakal (2025)	<i>Efficient Models Based on Deep Learning Technique for Indian Sign Language</i>	Uses landmark-based hand-pose information with transfer-learning models including VGG16, VGG19, ResNet50 and MobileNetV2 for ISL alphabet and numerical gesture recognition.	Validation accuracy: VGG16 99.67%, VGG19 99.16%, ResNet50 98.72%, MobileNetV2 99.69% . MobileNetV2 achieved the highest validation accuracy and smallest model size among the evaluated models. (ScienceDirect)	Focuses mainly on alphabetic and numeric gestures rather than continuous sign-language sentences and natural-language conversation.

Amira Elnashar et al. (2024)	<i>Bi-directional Translation Methods of ASL and Speech/Text</i>	Develops a bi-directional communication system supporting sign-to-speech/text through camera gestures or data gloves, and speech/text-to-sign through a 3D avatar.	Evaluates the system based on translation accuracy and demonstrates the potential for communication between sign-language and speech/text users. (ScienceDirect)	Uses ASL rather than ISL and incorporates multiple input/output technologies. The system does not primarily focus on contextual LLM-based conversation.
Ozge Mercanoglu Sincan et al. (2025)	<i>Gloss-Free Sign Language Translation: An Unbiased Evaluation of Progress in the Field</i>	Re-implements and compares recent gloss-free sign-language translation models using a unified codebase, standardized preprocessing, video encoders and training setups.	Provides a unified benchmark and comparative evaluation of gloss-free SLT models rather than proposing only one new model. It finds that some reported improvements decrease under standardized conditions. (ScienceDirect)	Highlights that results can depend heavily on backbone choice, preprocessing, training optimization and evaluation methodology , making direct comparison between studies difficult.

2.3 Research Gap Identification

Research Gap 1: Limited vocabulary and isolated-gesture focus. Many existing systems recognize predefined alphabets, words, or isolated gestures, which limits their ability to represent complete or continuous sign-language communication.

Research Gap 2: Difficulty with dynamic gestures and temporal variation. Video-based recognition can handle movement, but continuous signing requires sequence modelling and greater computational resources.

Research Gap 3: Environmental and user variation. Lighting, background, camera position, hand orientation, signing speed, and differences between users can affect recognition accuracy and generalization.

Research Gap 4: Real-time and accessibility constraints. Some approaches require larger datasets or higher computational resources, while others provide text only and do not support direct speech output.

Research Gap 5: Need for integrated communication output. The literature indicates value in combining gesture detection, classification, sign-to-text conversion, and sign-to-speech generation within a single accessible interface.

Based on these gaps, Beyond Words focuses on an integrated sign-language communication workflow that recognizes predefined gestures and converts them into understandable text and speech while maintaining a simple user interface.

CHAPTER 3

SYSTEM DESIGN & ARCHITECTURE

3.1 SYSTEM ARCHITECTURE

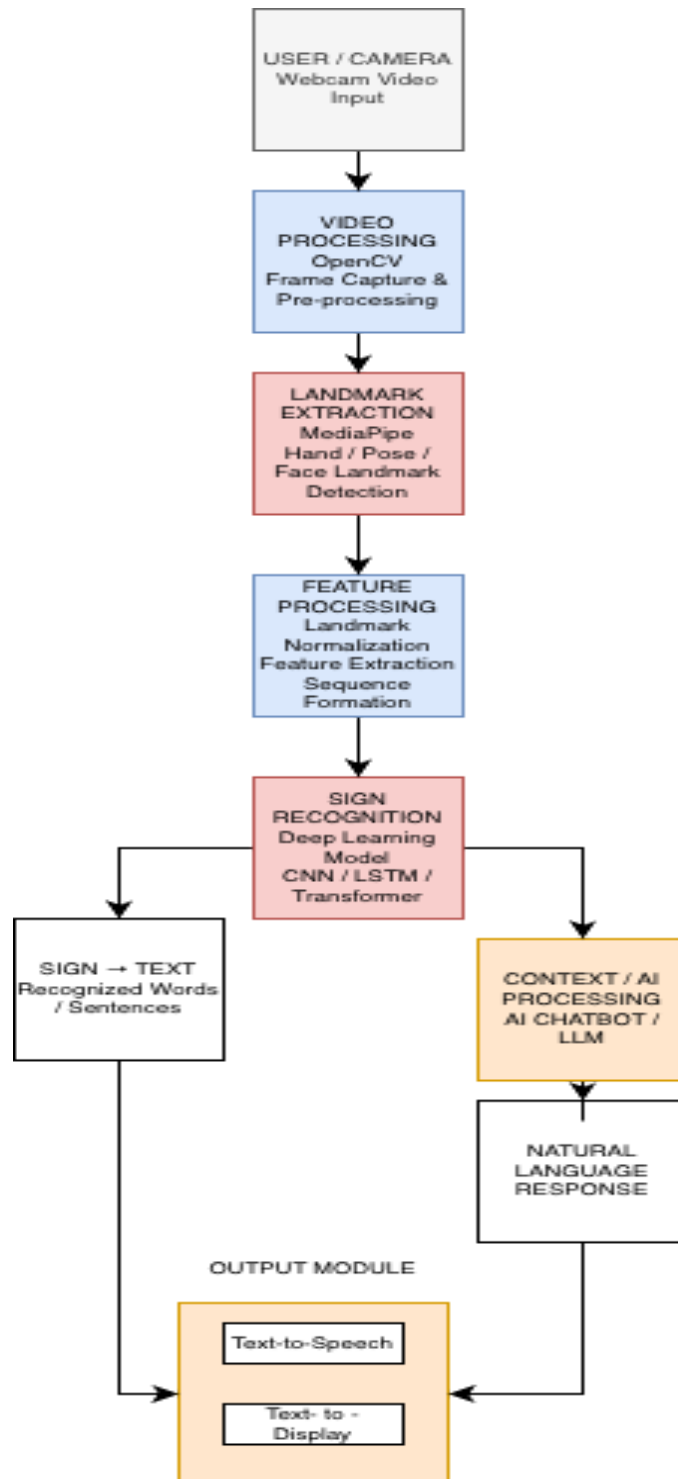


Fig 3.1

Figure 3.1 shows the flow from camera input through gesture detection and classification to text and speech output. The communication path is extended as camera input → preprocessing → hand detection and landmark extraction → gesture classification → sign interpretation → recognized text, with the recognized text routed to text output, text-to-speech, or the sign-language-driven AI chatbot. The chatbot processes the recognized text as a query and returns a response that can be displayed as text or converted to speech.

The architecture separates input, recognition, interpretation, and communication output into logical stages. The AI chatbot forms an additional interaction layer after sign-to-text conversion, enabling conversational responses based on the recognized sign-language input. The final stage provides understandable text and speech while supporting real-time interaction.

3.2 Module-Level Design

The **Beyond Words** system is divided into several interconnected modules: **Camera Input, Visual Processing, Gesture Detection, Gesture Classification, Gesture-to-Text Mapping, Text Output, Speech Processing, AI Chatbot, User Interface, and Accessibility**. Each module performs a specific function in the overall sign-language communication pipeline. The modules work together to capture sign-language gestures, extract meaningful visual features, recognize the intended signs, convert them into text, and generate spoken or conversational responses.

1. Camera Input Module

The Camera Input Module acts as the primary interface between the user and the Beyond Words system. It captures the user's hand gestures, body movements, and relevant facial expressions through a standard webcam or camera in real time.

The camera continuously captures a sequence of video frames rather than relying on individual photographs. These frames contain the visual information required to identify both static and dynamic sign-language gestures. The module manages camera initialization, frame acquisition, resolution settings, and communication with the visual processing pipeline.

A major objective of this module is to eliminate the need for specialized hardware. Users can interact with the system using a conventional laptop or external webcam, making the solution inexpensive and easier to deploy in educational institutions, homes, public spaces, and other real-world environments.

The captured frames are continuously transferred to the Visual Processing Module for preprocessing and analysis. The quality and consistency of the camera input directly affect the accuracy of subsequent landmark detection and gesture recognition.

2. Visual Processing Module

The Visual Processing Module prepares the raw camera frames for further analysis. **OpenCV** is used to handle camera communication, frame acquisition, image resizing, color-space conversion, and other image-processing operations.

Raw video frames may contain unnecessary information such as background objects, lighting variations, noise, and changes in camera distance. The preprocessing stage helps provide a consistent representation of the input before it is passed to the landmark detection system.

The module can perform operations such as resizing frames to a suitable resolution, converting color formats, improving consistency between frames, and displaying the processed camera feed. These operations reduce unnecessary computational overhead and allow the recognition pipeline to operate more efficiently.

The processed frames are then transferred to the Gesture Detection Module, where relevant hand, body, and facial landmarks are extracted.

3. Gesture Detection Module

The Gesture Detection Module is one of the core components of Beyond Words. It identifies important anatomical landmarks from the processed video frames using **MediaPipe**.

Depending on the requirements of the application, the module can detect landmarks associated with the hands, face, and body. Hand landmarks are particularly important because finger positions, hand orientation, and relative distances between joints contain significant information about sign-language gestures.

The detected landmarks are represented as numerical coordinates. Instead of processing the entire image directly, the system can use these coordinates as a compact representation of the user's movements. This reduces the amount of visual information that needs to be processed by the classification model.

For example, the system can identify the positions of the wrist, fingers, elbows, shoulders, and

other relevant points. These coordinates can subsequently be normalized to reduce the effect of differences in camera distance, user position, and hand size.

The module also supports the analysis of sequences of landmarks. This is particularly important for **dynamic signs**, where the meaning depends not only on the hand position but also on how the hand moves over time.

4. Gesture Classification Module

The Gesture Classification Module determines which sign is being performed based on the landmarks and features extracted from the user's movements. The numerical landmark data generated by MediaPipe is supplied to a machine-learning or deep-learning model. The model learns relationships between landmark patterns and predefined sign-language classes during the training stage.

For static signs, the model can analyze the spatial relationship between landmarks in a particular frame. For dynamic signs, multiple consecutive frames can be combined into a temporal sequence so that the model can recognize movement patterns. The classification process produces a predicted gesture along with, where supported, a confidence score. The confidence score can be used to determine whether the detected gesture is reliable enough to be converted into text.

This module allows the system to distinguish between visually similar gestures by considering the relative positions and movements of multiple landmarks rather than relying only on raw images.

5. Gesture-to-Text Mapping Module

The Gesture-to-Text Mapping Module converts the classified gesture labels into meaningful words, phrases, or sentences. The output of the classification model may initially be a technical class label such as a gesture identifier. The mapping module associates this identifier with its corresponding natural-language representation.\

For example, a recognized gesture can be mapped from an internal class label to a word such as **"Hello," "Thank you," or "I love you."** Multiple recognized signs can be combined sequentially to form longer messages. The module also handles the ordering and accumulation of recognized words. Instead of replacing the previous prediction every time a new gesture is detected, the system can maintain a temporary sentence buffer. This allows the user to construct a complete message through multiple signs.

Additional filtering mechanisms can be implemented to reduce duplicate predictions caused by the same gesture being detected across several consecutive frames. This improves the readability and stability of the generated text.

6. Text Output Module

The Text Output Module converts the recognized sign-language sequence into readable text and presents it to the user through the application's interface. Once a gesture has been classified and mapped to a word or phrase, the corresponding text is displayed in real time. This provides immediate visual feedback and allows individuals who do not understand sign language to interpret the user's intended message.

The module can maintain a history or sentence buffer containing previously recognized words. This allows users to construct longer messages instead of communicating one isolated gesture at a time. The generated text also serves as an important communication bridge between the sign-language recognition system and the AI Chatbot Module. The chatbot can receive the generated text as input and use it to understand the user's request or statement.

7. Speech Processing Module

The Speech Processing Module provides an audio-based communication mechanism for the recognized text. After the sign-language gestures are converted into text, the generated text can be passed to a **Text-to-Speech (TTS)** engine. The TTS system converts the textual output into natural-sounding speech. This enables a person who communicates primarily through sign language to communicate with a person who relies on spoken language without manually typing the message.

For example, when the user performs a recognized sign sequence corresponding to a sentence, the system first displays the sentence as text and can then convert the same sentence into audio. The speech module can include controls for starting, stopping, replaying, or adjusting the generated speech. This provides flexibility for users and makes the application more suitable for real-world communication scenarios.

8. AI Chatbot Module

The AI Chatbot Module provides the conversational intelligence of Beyond Words. Unlike a basic sign-to-text system that only translates gestures into written words, the chatbot allows the user to interact with the application through natural-language conversation.

The chatbot receives text generated by the Gesture Detection, Classification, and Text Output modules and treats it as a conversational input. **Natural Language Processing (NLP)** and artificial-intelligence techniques are used to understand the meaning, intent, and context of the user's message. The chatbot can generate an appropriate response based on the recognized input. The response can then be displayed as text within the application and optionally converted into speech through the Speech Processing Module.

This creates a two-way communication process:

Sign Language → Gesture Recognition → Text → AI Chatbot → Response Text → Speech

The chatbot therefore extends Beyond Words beyond simple gesture recognition and provides an interactive communication experience.

The module can also maintain conversational context so that responses are related to previous messages. For example, if the user asks a question through sign language and then performs another sign referring to the previous question, the chatbot can use the conversation history to provide a more meaningful response.

9. User Interface Module

The User Interface Module provides the visual environment through which users interact with Beyond Words. It integrates the camera feed, recognized gestures, generated text, chatbot conversation, and speech controls into a single application interface. The interface can display the live camera feed alongside detected landmarks, allowing users to understand whether the system is successfully detecting their gestures.

Recognized words and sentences can be displayed in a dedicated text area, while chatbot responses can be presented in a conversational chat interface. Buttons or controls can be provided for functions such as clearing the message, replaying speech, starting or stopping recognition, and interacting with the chatbot. The interface is designed to make the system understandable even for users who may not have technical knowledge.

10. Accessibility Module

The Accessibility Module focuses on making Beyond Words usable by people with different communication needs and levels of technical experience. The system is designed to minimize dependence on specialized equipment by using a standard camera as the primary input device. Visual feedback, readable text, and audio output provide multiple methods of receiving information.

The accessibility layer can also include adjustable text sizes, clear interface elements, high-contrast visual components, simple navigation, and audio feedback. These features can improve usability in educational institutions, public environments, and communication scenarios involving people with different abilities. The module ensures that accessibility is considered throughout the system rather than being treated as an additional feature.

11. Overall Module Interaction

The modules operate as a continuous processing pipeline. First, the **Camera Input Module** captures the user's video. The **Visual Processing Module** preprocesses the frames and forwards them to the **Gesture Detection Module**. MediaPipe extracts relevant landmarks, which are then passed to the **Gesture Classification Module**. The classification result is processed by the **Gesture-to-Text Mapping Module**, which converts the recognized gesture into a meaningful word or phrase.

The resulting text is displayed through the **Text Output and User Interface Modules**. At the same time, the text can be forwarded to the **AI Chatbot Module** for conversational processing. The chatbot generates a response that can be displayed as text and optionally converted into speech through the **Speech Processing Module**.

Therefore, the complete communication pipeline can be represented as:

Camera Input → Visual Processing → Landmark Detection → Gesture Classification → Gesture-to-Text Mapping → Text Output

For conversational interaction:

Text Output → AI Chatbot → Response Generation → Text Output → Text-to-Speech

This modular architecture allows each component to be independently developed, tested, improved, and replaced without requiring major changes to the entire system. It also provides a scalable foundation for expanding Beyond Words from a prototype supporting a limited set of signs into a more comprehensive real-time sign-language communication platform.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Data Collection and Preprocessing Methods

The system requires visual examples of predefined sign-language gestures for recognition. The report identifies the need for a suitable sign-language dataset containing different gestures and their corresponding labels. The data is used as the basis for gesture recognition and classification.

The captured or collected visual data is processed to identify relevant hand-gesture information before classification. Preprocessing is intended to reduce the effect of unnecessary visual variation and prepare consistent inputs for the recognition stage. Important gesture characteristics such as hand position, orientation, movement, and finger configuration can be used as recognition information. The literature reviewed in also identifies hand-landmark and keypoint approaches as useful methods for representing gesture information.

The preprocessing stage must also consider practical variations such as lighting conditions, background objects, camera position, hand orientation, distance from the camera, signing speed, and differences between users. These factors can affect recognition accuracy and therefore need to be considered during data preparation and system testing.

Finally, the prepared gesture information is supplied to the classification stage, where the detected sign is mapped to a predefined category and passed to the text and speech output stages.

4.2 Module Implementation Details

1. The proposed implementation begins with the camera input module, which captures the user's hand gesture. The visual processing and hand-detection stages prepare the input for recognition. After the sign is converted into text, the recognized text can be forwarded to the AI chatbot module as a conversational query.
2. The gesture classification module then applies machine-learning techniques to identify the predefined sign category. The recognized category is passed to the meaning-mapping stage so that the corresponding word or phrase can be produced.
3. The text output module displays the interpreted sign in a readable form. This output provides a direct visual representation of the recognized gesture and allows the receiving user to understand the message. The same recognized text can also be used as the input to the AI chatbot for conversational interaction.
4. The speech output module converts the recognized text into spoken output. This extends the system beyond screen-based communication and supports interaction with users who may not know sign language.
5. The real-time processing and user-interface stages coordinate recognition and output with minimal delay. The interface is intended to remain simple so that users can perform signs naturally in front of the camera.
6. The complete implementation follows the sequence camera input → gesture detection → sign classification → meaning mapping → recognized text → text output / speech output / AI chatbot query → chatbot response → text or speech output. This modular structure separates recognition from conversational response generation while maintaining an integrated communication workflow.
7. The report describes the proposed workflow and functional requirements but does not provide final implementation metrics or a completed performance evaluation.

4.3 Results and Discussion

The proposed system is designed to recognize predefined sign-language gestures from camera input and convert the recognized signs into understandable text or speech. The chatbot extension additionally enables recognized sign-language text to be used as a conversational query, with a generated response returned as text and optionally speech.

The literature review supports computer-vision, hand-landmark, machine-learning, deep-learning, and temporal approaches while identifying environmental and signer variation as important challenges.

Beyond Words integrates recognition with text and speech output to provide an accessible communication workflow for users who do not understand sign language. It also extends the workflow with an AI chatbot that accepts recognized sign-language text and generates conversational responses, providing an interactive communication path beyond direct translation.

4.4 Output Screenshots

The figures below are illustrative representations of the intended Beyond Words interface and output flow.



Figure 4.1 – Recognized Sign and Text Output

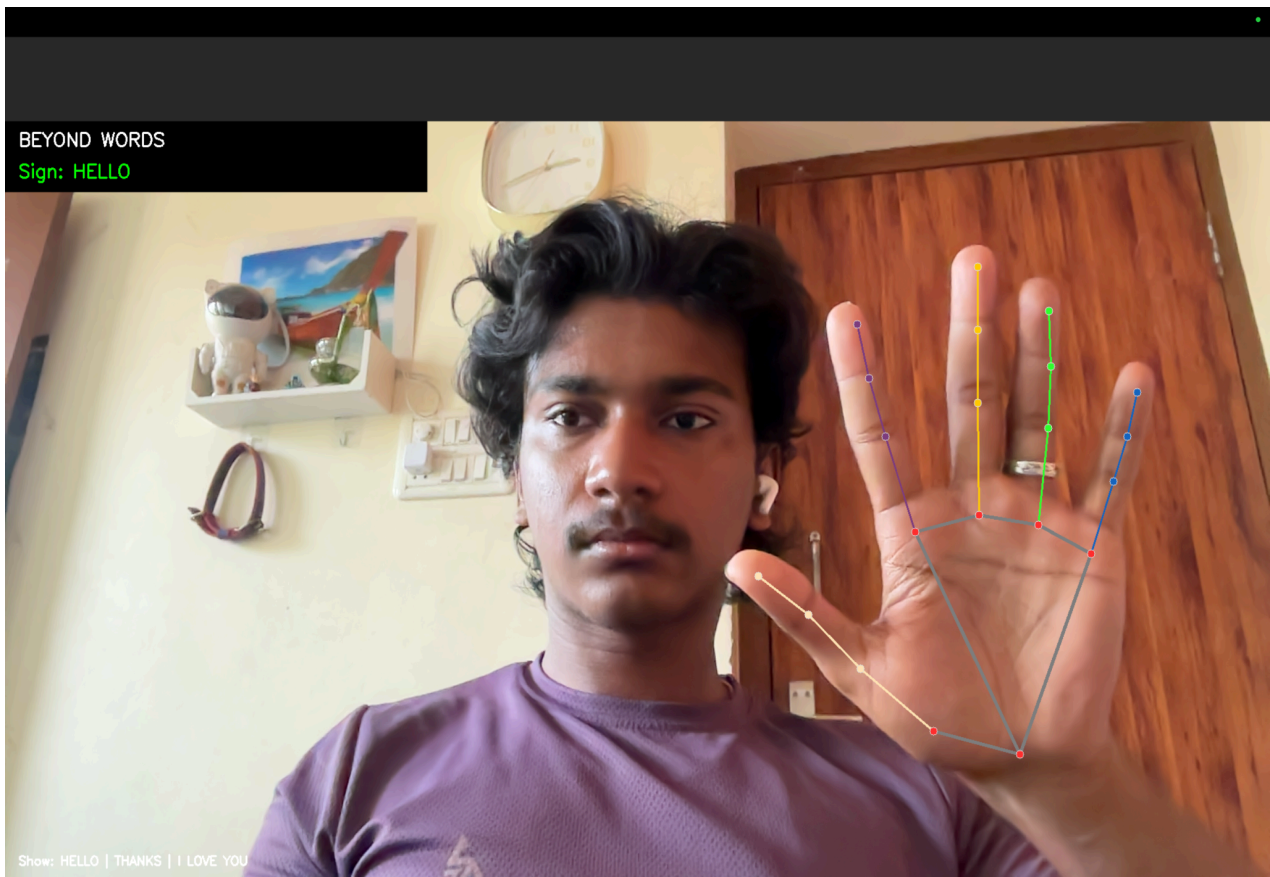


Fig 4.2 – Sign-to-Speech Output

CHAPTER 5

CONCLUSION

Beyond Words is proposed as an AI-based sign-language communication system that recognizes predefined hand gestures from visual input and converts them into understandable text or speech. The project combines computer vision, machine-learning-based gesture classification, meaning mapping, and accessible communication output in a single workflow. The system is further extended with a sign-language-driven AI chatbot that uses recognized text as conversational input and returns responses as text or speech.

The literature review shows that existing approaches provide a strong foundation for sign-language recognition but continue to face challenges involving limited vocabulary, dynamic gestures, environmental variation, signer diversity, computational requirements, and output accessibility. Beyond Words addresses these concerns at the system-design level by integrating recognition with both text and speech output. Overall, the project demonstrates the potential of Artificial Intelligence, Computer Vision, and Machine Learning to support more inclusive communication.

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APPENDIX

```
import cv2
import mediapipe as mp
import math
import pyttsx3

mp_hands = mp.solutions.hands
mp_drawing = mp.solutions.drawing_utils
mp_drawing_styles = mp.solutions.drawing_styles

hands = mp_hands.Hands(
    static_image_mode=False,
    max_num_hands=2,
    min_detection_confidence=0.6,
    min_tracking_confidence=0.6
)

# Text-to-speech
engine = pyttsx3.init()
engine.setProperty("rate", 150)

cap = cv2.VideoCapture(1)

def distance(p1, p2):
    return math.sqrt(
        (p1.x - p2.x) ** 2 +
        (p1.y - p2.y) ** 2
    )

def fingers_up(hand):
    """
    Returns:
```

```
[thumb, index, middle, ring, pinky]
```

```
1 = finger extended
```

```
0 = finger folded
```

```
"""
```

```
fingers = []
```

```
# Thumb
```

```
if hand.landmark[4].x < hand.landmark[3].x:
```

```
    fingers.append(1)
```

```
else:
```

```
    fingers.append(0)
```

```
# Index
```

```
if hand.landmark[8].y < hand.landmark[6].y:
```

```
    fingers.append(1)
```

```
else:
```

```
    fingers.append(0)
```

```
# Middle
```

```
if hand.landmark[12].y < hand.landmark[10].y:
```

```
    fingers.append(1)
```

```
else:
```

```
    fingers.append(0)
```

```
# Ring
```

```
if hand.landmark[16].y < hand.landmark[14].y:
```

```
    fingers.append(1)
```

```
else:
```

```
    fingers.append(0)
```

```
# Pinky
```

```
if hand.landmark[20].y < hand.landmark[18].y:
```

```
    fingers.append(1)
else:
    fingers.append(0)

return fingers
```

```
def recognize_sign(hand_landmarks):

    fingers = fingers_up(hand_landmarks)

    thumb, index, middle, ring, pinky = fingers

    if (
        thumb == 1 and
        index == 1 and
        middle == 0 and
        ring == 0 and
        pinky == 1
    ):
        return "I LOVE YOU"

    if (
        thumb == 1 and
        index == 1 and
        middle == 1 and
        ring == 1 and
        pinky == 1
    ):
        return "HELLO"

    if (
        thumb == 1 and
```

```

        index == 1 and
        middle == 0 and
        ring == 0 and
        pinky == 0
    ):
        return "THANKS"

    return "UNKNOWN"

last_sign = ""
spoken_sign = ""

while cap.isOpened():

    success, frame = cap.read()

    if not success:
        print("Could not read camera")
        break

    # Flip camera for mirror effect
    frame = cv2.flip(frame, 1)

    # Convert BGR → RGB
    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    # Process hand
    results = hands.process(rgb_frame)

    detected_sign = "UNKNOWN"
    if results.multi_hand_landmarks:

        for hand_landmarks in results.multi_hand_landmarks:

```

```

# Draw all 21 hand points
mp_drawing.draw_landmarks(
    frame,
    hand_landmarks,
    mp_hands.HAND_CONNECTIONS,
    mp_drawing_styles.get_default_hand_landmarks_style(),
    mp_drawing_styles.get_default_hand_connections_style()
)

# Recognize sign
detected_sign = recognize_sign(hand_landmarks)

cv2.rectangle(
    frame,
    (0, 0),
    (640, 100),
    (0, 0, 0),
    -1
)

cv2.putText(
    frame,
    "BEYOND WORDS",
    (20, 35),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.9,
    (255, 255, 255),
    2
)

cv2.putText(
    frame,
    "Sign: " + detected_sign,
    (20, 80),
    cv2.FONT_HERSHEY_SIMPLEX,

```

```

    0.9,
    (0, 255, 0),
    2
)
if detected_sign != "UNKNOWN":

    if detected_sign != spoken_sign:

        engine.say(detected_sign)
        engine.runAndWait()

        spoken_sign = detected_sign

    else:
        spoken_sign = ""

cv2.putText(
    frame,
    "Show: HELLO | THANKS | I LOVE YOU",
    (20, frame.shape[0] - 20),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.6,
    (255, 255, 255),
    2
)

cv2.imshow("Beyond Words - Sign Language Recognition", frame)

# Press Q to quit
if cv2.waitKey(1) & 0xFF == ord("q"):
    break

cap.release()

```

```
cv2.destroyAllWindows()
```

```
hands.close()
```